

PROJECT REPORT

Stock Market Trend Prediction Using Machine Learning

Domain	Finance
Stock Symbol	AAPL (Apple Inc.)
Data Source	Yahoo Finance (yfinance)
Period	January 2018 – December 2024
Algorithm	Random Forest Classifier
Report Date	May 29, 2026

This report presents a complete end-to-end data science project that builds a machine learning model to predict next-day stock price direction (UP / DOWN) for Apple Inc. (AAPL). The project encompasses data collection, exploratory data analysis, technical indicator engineering, model training with a Random Forest Classifier, performance evaluation, and actionable conclusions for future improvement.

1. Project Overview

The ability to predict short-term stock movements is a classic problem in quantitative finance. This project applies a supervised machine learning approach—specifically, a Random Forest Classifier—to classify whether the next-day closing price of AAPL stock will be higher (UP) or lower (DOWN) than the current day.

Objectives

- Collect and prepare seven years of real AAPL stock data from Yahoo Finance.
- Engineer 13+ technical indicators commonly used by traders and analysts.
- Build and train a Random Forest Classifier with appropriate time-series splitting.
- Evaluate model performance using accuracy, precision, recall, and F1-score.
- Identify the most predictive features and propose improvements.

Technology Stack

Library / Tool	Purpose
Python 3	Core programming language
yfinance	Download historical OHLCV stock data
pandas / numpy	Data manipulation and numerical computation
scikit-learn	Machine learning (Random Forest, metrics)
matplotlib / seaborn	Data visualisation
ta (Technical Analysis)	RSI, MACD, Bollinger Bands computation

2. Dataset Description

Historical daily OHLCV (Open, High, Low, Close, Volume) data for AAPL was downloaded from Yahoo Finance covering 2018-01-02 to 2024-12-31. After feature engineering and removal of rows with NaN values (due to rolling windows), the final dataset contains 1,761 trading days.

Raw Dataset Sample (First 5 Rows)

Date	Close (\$)	High (\$)	Low (\$)	Open (\$)	Volume
2018-01-02	40.27	40.28	39.57	39.78	102,223,600
2018-01-03	40.26	40.80	40.20	40.33	118,071,600
2018-01-04	40.45	40.55	40.23	40.33	89,738,400
2018-01-05	40.91	40.99	40.45	40.54	94,640,000
2018-01-08	40.76	41.05	40.66	40.76	82,271,200

Data Quality

A missing-value check confirmed zero null entries across all six raw columns (Date, Close, High, Low, Open, Volume), ensuring a clean baseline for feature engineering.

Key Descriptive Statistics

Feature	Mean	Std Dev	Min	Max
Close (\$)	123.19	58.46	33.74	257.38
High (\$)	124.41	58.92	34.57	258.45
Low (\$)	121.83	57.91	33.69	255.99
Volume	98.96M	54.27M	23.23M	426.51M
Daily Return	0.12%	1.93%	-12.86%	11.98%
RSI	55.62	12.69	22.22	85.72
MACD	0.82	2.44	-6.48	8.88
Volatility (5d)	0.017	0.011	0.002	0.107

3. Feature Engineering

Thirteen technical indicators were derived from the raw OHLCV data. These indicators are widely used in technical analysis to capture trend, momentum, volatility, and volume dynamics.

Feature	Category	Description
Daily_Return	Return	Percentage change in closing price
SMA_10	Trend	10-day Simple Moving Average of Close
SMA_20	Trend	20-day Simple Moving Average of Close
EMA_10	Trend	10-day Exponential Moving Average
RSI	Momentum	Relative Strength Index (14-day window)
MACD	Momentum	Moving Average Convergence Divergence
MACD_Signal	Momentum	9-day EMA of MACD line
BB_High	Volatility	Bollinger Band upper limit (20d, 2 sigma)
BB_Low	Volatility	Bollinger Band lower limit (20d, 2 sigma)
Volatility_5	Volatility	5-day rolling standard deviation of returns
Price_Change	Return	Absolute daily change in Close price
High_Low_Spread	Range	Intra-day spread: High minus Low
Volume_Change	Volume	Percentage change in trading volume

Target Variable

The binary target variable is defined as: **Target = 1 (UP)** if tomorrow's closing price is higher than today's, else **Target = 0 (DOWN)**. Approximately 53.82% of trading days resulted in an upward next-day movement, indicating a slight class imbalance favoring UP days.

4. Methodology

Train / Test Split

For time-series data, random shuffling must be avoided to prevent look-ahead bias. A chronological 80/20 split was applied: the first 1,382 samples (2018–2023) form the training set, while the last 346 samples (2024) serve as the hold-out test set.

Split	Samples	Period	Proportion
Training	1,382	Jan 2018 – ~Oct 2023	80%
Testing	346	~Oct 2023 – Dec 2024	20%
Total	1,728	Jan 2018 – Dec 2024	100%

Model: Random Forest Classifier

A Random Forest is an ensemble of decision trees trained on bootstrapped subsets of data with random feature selection at each split. It is well-suited for this task because it handles non-linear relationships, is robust to overfitting, and provides built-in feature importance scores.

Hyperparameters Used

Parameter	Value	Rationale
n_estimators	200	Large ensemble reduces variance
max_depth	8	Prevents deep overfitting trees
min_samples_split	10	Requires sufficient data to split a node
min_samples_leaf	4	Ensures each leaf has meaningful support
random_state	42	Reproducibility

5. Model Evaluation Results

Overall Accuracy

Model Accuracy	45.38%	(Baseline: ~53.8%)
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Classification Report

Class	Precision	Recall	F1-Score	Support
0 (DOWN)	0.44	0.97	0.60	149
1 (UP)	0.72	0.07	0.12	197
Macro Avg	0.58	0.52	0.36	346
Weighted Avg	0.60	0.45	0.33	346

Confusion Matrix Analysis

The confusion matrix reveals the model's strong bias toward predicting DOWN movements (class 0). While it achieves 97% recall for DOWN days—capturing nearly all downturns—it severely under-predicts UP days (only 7% recall for class 1). This imbalance suggests the model learned a conservative, risk-averse pattern, which may be partly driven by the class distribution and hyperparameter settings.

	Predicted: DOWN	Predicted: UP
Actual: DOWN	145 (TP)	4 (FN)
Actual: UP	183 (FP)	14 (TN)

Feature Importance (Top 10)

Rank	Feature	Importance Score	Category
1	Daily_Return	0.0838	Return
2	RSI	0.0793	Momentum
3	Volatility_5	0.0717	Volatility
4	Price_Change	0.0686	Return
5	High_Low_Spread	0.0630	Range
6	MACD_Signal	0.0621	Momentum
7	MACD	0.0615	Momentum
8	Volume_Change	0.0615	Volume
9	Volume	0.0540	Volume

10	BB_Low	0.0496	Volatility
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Daily Return and RSI are the top predictors, underscoring the importance of momentum signals. Volatility (5-day rolling std dev) ranks third, reflecting that high-volatility periods contain stronger directional signals.

6. Sample Predictions

The table below shows 10 consecutive predictions from the test set, illustrating the model's tendency to predict DOWN in the majority of cases.

Index	Actual	Predicted	Correct?
1415	UP	DOWN	✗
1416	UP	UP	✓
1417	UP	DOWN	✗
1418	UP	DOWN	✗
1419	DOWN	DOWN	✓
1420	UP	DOWN	✗
1421	UP	DOWN	✗
1422	UP	DOWN	✗
1423	UP	DOWN	✗
1424	UP	DOWN	✗

7. Next-Day Prediction (Latest Signal)

The trained model was applied to the most recent available trading day's features to generate a forward-looking signal for the next trading session.

Signal	Probability of DOWN	Probability of UP
DOWN ↓	62.89%	37.11%

Disclaimer: This prediction is for academic/educational purposes only and should not be used as investment advice.

8. Conclusions

Complete Workflow Demonstrated

The project successfully implemented a full data science pipeline: data acquisition, EDA, feature engineering, modelling, evaluation, and prediction.

Model Accuracy Below Baseline

The Random Forest achieved 45.38% accuracy, slightly below the naive baseline of ~53.8% (always predicting UP). This reveals the model learned a DOWN-biased strategy rather than a balanced classifier.

Class Imbalance Impact

The mild imbalance (54% UP vs 46% DOWN) combined with conservative hyperparameters led to high recall for DOWN (97%) at the expense of UP recall (7%). Techniques like SMOTE or `class_weight='balanced'` are recommended.

Feature Relevance Confirmed

Daily Return, RSI, and short-term Volatility are the three most predictive features, validating the inclusion of momentum and volatility indicators.

Practical Utility

Even with modest accuracy, such a model can be part of a larger trading system where it contributes a risk-adjusted signal alongside other filters.

9. Future Improvements

- Expand to a portfolio of stocks (S&P; 500 universe) to test generalisability.
- Incorporate news sentiment analysis via NLP to add exogenous signals.
- Try gradient boosting models (XGBoost, LightGBM) and LSTM neural networks for sequence modelling.
- Apply hyperparameter optimisation (GridSearchCV, Optuna) for better precision/recall trade-offs.
- Address class imbalance with SMOTE oversampling or `class_weight='balanced'`.
- Backtest trading strategies using predicted signals with transaction cost simulation.
- Add macro-economic features: interest rates, VIX, sector ETF performance.